PUSAT JANTUNG Nasional
National Cardiovescular Center Harapan Kita

Transesophageal Echocardiography Report

UNIT PELAKSANA FUNGSIONAL (UPF) DIAGNOSTIK NON INVASIF
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SR Recovery

ID : Fellow1

TEE Procedure

Anesthesia	Sedation	Duration	Complication
Topical spray xyloce 10%	No	15 min	None

Patient Information

Name SUHAICIH ORIN	**Study Date** 03/01/2024	**Height** 155 cm	**Diagnosis** HOCM, MR moderate
MRN 2023540463	**DOB** 17/07/1964	**Weight** 89 kg	**Reffering physician** dr. Dwita Ryan, SpJP
Age	**Gender** female	**BSA** 1.87 m^2	**Operator** dr. Estu Rudiktyo, SpJP (K)
First Write Time 03/01/2024 16:14:06	**Last Write Time** 05/01/2024 15:18:04	**Blood Pressure** 163 / 88 mmHg	**Medical Device** Vivid E95

MMode / 2D Measurements & Calculations

IVSd 13.6 mm	**Aor Diam** 36.6 mm	**EDV(teich)** 113.5 ml	**EF(Simpson's Biplane)** 55 %
IVSs 27 mm	**LA dimen** 44.6 mm	**ESV(teich)** 12 ml	**EF(Simpson's 4ch)** %
LVIDd 49.1 mm	**LVOT diam** mm	**EF(teich)** 89 %	**EF(Simpson's 2ch)** %
LVIDs 19.8 mm	**LA/Ao** 1.2	**FS** 59.7 %	**IVC(Ins)** mm
LVPWd 22.9 mm	**LAVI** 39.07 ml/m^2	**TAPSE** 26 mm	**IVC(EXP)** mm
LVPWs 29 mm	**LV mass Index** 226 g/m^2		

Doppler Measurements & Calculations

MV E vel 90 m/sec	**E/E'(medial)** 15.61	**MV V max** m/sec	**MV VTI** cm
MV DT 227 msec	**E/E'(lateral)** 15.11	**MV mean PG** mmHg	**MV PHT** msec
MV A vel 78 m/sec	**E/E'(average)** 15.36	**MVA(plan)** cm^2	**MVA(VTI)** cm^2
E/A 1.15			**MVA by PHT** cm^2

AV V max m/sec	**AI PHT** msec	**TR V max(systolic)** m/sec
AV max PG mmHg	**AI(EROA)** cm^2	**TR max PG(systolic)** mmHg
AV mean PG mmHg	**LVOT VTI** cm	**TV mean PG(diastolic)** mmHg
AV VTI cm	**Desending Aorta(EDV)** m/sec	**TV VTI** cm
AVA (VTI) cm^2	**SV(LVOT)** ml	**PV Acc time** msec
AVA(Vmax) cm^2	**AVA(plan)** cm^2	

Findings

[INSERT]

ECG Rhythm

- [x] [E01] ECG is sinus rhythm.
- [ ] [E02] ECG is in atrial fibrillation.
- [ ] [E03] ECG is pacing rhythm.
- [ ] [E04] ECG is

ECG is sinus rhythm.

[E05] Add Statement

- [ ]

Result

ECG is sinus rhythm.

Study Quality

- [ ] [S01] The quality of the study is good.
- [x] [S02] The quality of the study is adequate.
- [ ] [S03] The quality of the study is poor.
- [ ] [S04] The study is technically difficult due to

The quality of the study is adequate.

[S05] Add Statement

- [ ]

Result

The quality of the study is adequate.

LV

- [ ] [LV01] Normal

Cavity Size

- [x] [LVCS01] Left ventricle size is normal.
- [ ] [LVCS02] Left ventricle is dilated.
- [ ] [LVCS03] Left ventricle is smallish.
- [ ] [LVCS04] Left ventricle is D-shaped.

Left ventricle size is normal.

Wall Thickness

- [ ] [LVWT01] The wall thickness is normal.
- [ ] [LVWT02] There is concentric remodeling of left ventricle.
- [x] [LVWT03] There is concentric left ventricular hypertrophy.
- [ ] [LVWT04] There is eccentric left ventricular hypertrophy.

There is asymmetric septal hypertrophy. RWT 0.9 ; LVMI 226 gr/m2. There is dyn

Global Funtion

- [x] [LVGF01] Global left ventricular systolic function is normal.
- [ ] [LVGF02] Global left ventricular systolic function is reduced.

Global left ventricular systolic function is normal.

Diastolic Function

- [x] [LVDF01] LV Diastolic Function is normal.
- [ ] [LVDF02] LV Diastolic Function is abnormal.

LV Diastolic Function is indeterminate

[LVAS01] Add Statement

- [x]

Result

Left ventricle size is normal. There is asymmetric septal hypertrophy. RWT 0.9 ; LVMI 226 gr/m2. There is dynamic gradient at LVOT 68-95 mmHg. Global left ventricular systolic function is normal. LV Diastolic Function is indeterminate

RV

- [x] [RV01] Normal

Right ventricle is normal in size and function.

RV Size

- [ ] [RVS01] The right ventricle is normal in size.

☐ [RVS01] The right ventricle is normal in size.
☐ [RVS02] The right ventricle is dilated

RV Wall Thickness

☐ [RVWT01] Wall thickness of the right ventricle is normal.
☐ [RVWT02] There is right ventricular hypertrophy.

RV Systolic Function

☐ [RVSF01] Right ventricular systolic function is normal.
☐ [RVSF02] Right ventricular systolic function is decreased.

[RVAS01] Add Statement

☐

Result

Right ventricle is normal in size and function.

LA

LA Size

☐ [LAS01] Left atrial size is normal.
☑ [LAS02] Left atrium is dilated.

Left atrium is dilated.

[LAAS01] Add Statement

☐

Result

Left atrium is dilated.

RA

RA Size

☑ [RAS01] Right atrial size is normal.
☐ [RAS02] Right atrium is dilated.

Right atrial size is normal.

[RAAS01] Add Statement

☐

Result

Right atrial size is normal.

Interatrial Septum

☑ [INTERSEP01] Intact interatrial septum.
☐ [INTERSEP02] There is atrial septal aneurysm.
☐ [INTERSEP03] Hyperkinetic interatrial septum.
☐ [INTERSEP04] There is gap at interatrial septum with diameter of
☐ [INTERSEP05] There is stretched PFO L to R shunt .
☐ [INTERSEP06] There is iatrogenic ASD with diameter of
☐ [INTERSEP07] Post ASD closure.
☐ [INTERSEP08] Post device closure.

Intact interatrial septum.

[INTERSEP09] Add Statement

☐

Result

Intact interatrial septum.

Interventricular Septum

☑ [INTERVSEP01] Intact interventricular septum.
☐ [INTERVSEP02] There is gap at IVS, with diameter of
☐ [INTERVSEP03] There is ventricular septal rupture.

Intact interventricular septum.

[INTERVSEP04] Add Statement

☐

Result

Intact interventricular septum.

Aortic Valve

☐ [AV01] Normal

AV Structure List

☐ [AVSL01] The aortic valve is not well visualized, number cusp could not be identified.
☑ [AVSL02] The aortic valve is trileaflet.
☐ [AVSL03] The aortic valve is bicuspid.
☐ [AVSL04] The aortic valve is
☐ [AVSL05] The aortic valve leaflets appear to be thickened.
☐ [AVSL06] The aortic valve appears rheumatic
☐ [AVSL07] There is no aortic valve calcification.
☐ [AVSL08] The cusp is heavily calcified.
☑ [AVSL09] There is calcification at.
☐ [AVSL10] There is nodular thickening of

The aortic valve is trileaflet.

There is calcification at LCC.

AV Function

☑ [AVF01] The aortic valve opens well.
☐ [AVF02] The aortic valve is not heavily calcified.
☐ [AVF03] There is systolic doming of aortic valve.
☐ [AVF04] There is prolapse of
☐ [AVF05] There is flail of.

The aortic valve opens well.

AV Regurgitation List

☑ [AVRL01] No aortic regurgitation is seen.
☐ [AVRL02] There is trace aortic regurgitation.
☐ [AVRL03] There is mild aortic regurgitation.
☐ [AVRL04] There is mild-moderate aortic regurgitation.
☐ [AVRL05] There is moderate aortic regurgitation.
☐ [AVRL06] There is moderate-severe aortic regurgitation.
☐ [AVRL07] There is severe aortic regurgitation.
☐ [AVRL08] Regurgitation jet is central.
☐ [AVRL09] Regurgitation jet is eccentric.

No aortic regurgitation is seen.

AV Stenosis List

☑ [AVSTL01] There is no valvular aortic stenosis.
☐ [AVSTL02] There is mild valvular aortic stenosis.
☐ [AVSTL03] There is mild-moderate valvular aortic stenosis.
☐ [AVSTL04] There is moderate valvular aortic stenosis.
☐ [AVSTL05] There is moderate-severe valvular aortic stenosis.
☐ [AVSTL06] There is severe valvular aortic stenosis.
☐ [AVSTL07] There is subaortic stenosis.
☐ [AVSTL08] There is supravalvar aortic stenosis.
☐ [AVSTL09] There is paradoxical aortic stenosis.
☐ [AVSTL10] There is low flow low gradient aortic stenosis.

There is no valvular aortic stenosis.

AV Vegetations & Abcess List

☐ [AVVA01] No vegetation is seen.
☐ [AVVA02] We cannot exclude vegetation in the aortic valve.
☐ [AVVA03] There is vegetation at.
☐ [AVVA04] There is mobile vegetation at
☐ [AVVA05] There is abcess at

AV Prosthetic List

☐ [AVPL01] Aortic valve post repair.
☐ [AVPL02] Aortic valve post replace with bioprosthetic.
☐ [AVPL03] Aortic valve post replace with mechanical valve.
☐ [AVPL04] A homograft in aortic valve is in present.
☐ [AVPL05] The opening prosthetic valve is not well visualized.
☐ [AVPL06] The prosthetic valve is well-seated.
☐ [AVPL07] The prosthetic valve opens well.
☐ [AVPL08] There is limited prosthetic valve opening

☐ [AVPL09] There is no thrombus see on prosthetic valve.
☐ [AVPL10] There is possible thrombus on prosthetic valve.
☐ [AVPL11] No leakage is seen.
☐ [AVPL12] There is …… central leakage.
☐ [AVPL13] There is …… paravalvar leakage.
☐ [AVPL14] Doppler evaluation showed normal prosthetic aortic valve gradient.
☐ [AVPL15] Doppler evaluation showed increased prosthetic aortic valve gradient.
☐ [AVPL16] The prosthetic vegetations are suspected.
☐ [AVPL17] The prosthetic vegetations are present.
☐ [AVPL18] The rocking suggests dehiscence in the aortic valve.

[AVAS01] Add Statement

☐

Result

The aortic valve is trileaflet. There is calcification at LCC. The aortic valve opens well. No aortic regurgitation is seen. There is no valvular aortic stenosis.

Mitral Valve

☐ [MV01] Normal

MV Structure List

☐ [MVSL01] The mitral valve is not well visualized.
☑ [MVSL02] The mitral valve structure is normal.
☐ [MVSL03] The mitral valve leaflets appear to be thickened.
☐ [MVSL04] The mitral valve appears rheumatic.
☐ [MVSL05] Mitral valve leaflets appear myxomatous.
☐ [MVSL06] There is redundant elongated chordae.
☐ [MVSL07] There is no annular calcification.
☐ [MVSL08] There is mitral annular calcification.
☐ [MVSL09] Mitral cleft is present.
☐ [MVSL10] Valve chordae is thickened and calcified.
☐ [MVSL11] Papillary muscle is thickened and calcified.

The mitral valve structure is normal.

MV Function List

☑ [MVFL01] Mitral valve opens well.
☐ [MVFL02] There is doming of the mitral valve with minimal stenosis.
☐ [MVFL03] There is doming of the mitral valve with restricted opening.
☐ [MVFL04] Opening of mitral valve is restricted.
☐ [MVFL05] There is restricted of the mitral leaflet (systolic tenting)
☐ [MVFL06] There is billowing of mitral valve closure, with no prolapse.
☐ [MVFL07] There is prolapse of
☐ [MVFL08] There is flail of.
☐ [MVFL09] Chordal rupture is seen.
☐ [MVFL10] There is perforation of.
☐ [MVFL11] There is no SAM is seen.
☐ [MVFL12] There is chordal SAM.
☐ [MVFL13] Systolic anterior motion is present.

Mitral valve opens well.

MV Regurgitation List

☐ [MVRL01] No mitral regurgitation is seen.
☐ [MVRL02] There is trace mitral regurgitation.
☐ [MVRL03] There is mild mitral regurgitation.
☐ [MVRL04] There is mild-moderate mitral regurgitation.
☐ [MVRL05] There is moderate mitral regurgitation.
☐ [MVRL06] There is moderate-severe mitral regurgitation.
☐ [MVRL07] There is severe mitral regurgitation.
☐ [MVRL08] Regurgitation jet is central.
☐ [MVRL09] Regurgitation jet is eccentric.
☐ [MVRL10] Pulmonary vein systolic reversal flow is present.

MV Stenosis List

☐ [MVSTL01] There is no mitral stenosis.
☐ [MVSTL02] There is mild mitral stenosis.
☐ [MVSTL03] There is mild-moderate mitral stenosis.
☐ [MVSTL04] There is moderate mitral stenosis.
☐ [MVSTL05] There is moderate-severe mitral stenosis.
☐ [MVSTL06] There is severe mitral stenosis.
☐ [MVSTL07] Wilkin's score.

MV Vegetations & Abcess List

☐ [MVVA01] No vegetation is seen.
☐ [MVVA02] We cannot exclude vegetation at the mitral valve.
☐ [MVVA03] There is vegetation at.
☐ [MVVA04] There is mobile vegetation at
☐ [MVVA05] There is abcess at

MV Repair / Prosthetic List

☐ [MVRPL01] Mitral valve post repair.
☐ [MVRPL02] Mitral valve post repair with ring annuloplasty.
☐ [MVRPL03] Mitral valve post replace with bioprosthetic.
☐ [MVRPL04] Mitral valve post replace with mechanical valve.
☐ [MVRPL05] There is no residual mitral regurgitation.
☐ [MVRPL06] There is …. residual mitral regurgitation.
☐ [MVRPL07] There is … residual mitral stenosis.
☐ [MVRPL08] Prosthetic valve is abnormal.
☐ [MVRPL09] The opening prosthetic valve is not well visualized.
☐ [MVRPL10] The prosthetic valve is well-seated.
☐ [MVRPL11] The prosthetic valve opens well.
☐ [MVRPL12] There is limited prosthetic valve opening.
☐ [MVRPL13] There is no thrombus see on prosthetic valve.
☐ [MVRPL14] There is possible thrombus on prosthetic valve.
☐ [MVRPL15] No leakage is seen.
☐ [MVRPL16] There is …… central leakage.
☐ [MVRPL17] There is …… paravalvar leakage.
☐ [MVRPL18] Doppler evaluation showed normal prosthetic mitral valve gradient.
☐ [MVRPL19] Doppler evaluation showed increased prosthetic mitral valve gradient.
☐ [MVRPL20] There is abnormal prosthetic MV gradient.
☐ [MVRPL21] The prosthetic vegetations are suspected.
☐ [MVRPL22] The prosthetic vegetations are present.
☐ [MVRPL23] The rocking suggests dehiscence in the mitral valve.

[MVAS01] Add Statement

☑

There is moderate mitral regurgitation due to systolic anterior motion of the AML (disrupting the normal coaptation of AML and PML). MR VCD 3.4 mm. Blunted S wave at pulmonary vein flow. There is no mitral stenosis.

Result

The mitral valve structure is normal. Mitral valve opens well. There is moderate mitral regurgitation due to systolic anterior motion of the AML (disrupting the normal coaptation of AML and PML). MR VCD 3.4 mm. Blunted S wave at pulmonary vein flow. There is no mitral stenosis.

Tricuspid Valve

☑ [TV01] Normal

Tricuspid valve is normal in structure and function. There is no tricuspid regurgitation.

TV Structure List

☐ [TVSL01] The tricuspid valve is not well visualized.
☐ [TVSL02] The tricuspid valve structure is normal.
☐ [TVSL03] The tricuspid valve appeared rheumatic.
☐ [TVSL04] Tricuspid leaflets are thickened.
☐ [TVSL05] The valvular appearance is consistent with carcinoid heart disease.
☐ [TVSL06] There is no tricuspid annular calcification.
☐ [TVSL07] There is tricuspid annular calcification.

TV Function List

☐ [TVFL01] Tricuspid valve opens well.
☐ [TVFL02] There is doming of tricuspid valve with minimal stenosis.
☐ [TVFL03] There is doming of the triscuspid valve with restricted opening.
☐ [TVFL04] Tenting tricuspid leaflet.
☐ [TVFL05] Uncoaptation of tricuspid closure is seen.
☐ [TVFL06] There is prolapse of
☐ [TVFL07] There is flail of

TV Regurgitation List

☐ [TVRL01] There is no tricuspid regurgitation.
☐ [TVRL02] There is trace tricuspid regurgitation.
☐ [TVRL03] There is mild tricuspid regurgitation.
☐ [TVRL04] There is mild-moderate tricuspid regurgitation.
☐ [TVRL05] There is moderate tricuspid regurgitation.
☐ [TVRL06] There is moderate-severe tricuspid regurgitation.
☐ [TVRL07] There is severe tricuspid regurgitation.
☐ [TVRL08] Regurgitation jet is central.
☐ [TVRL09] Regurgitation jet is eccentric.
☐ [TVRL10] Hepatic vein systolic reversal flow is seen.

TV Stenosis List

☐ [TVSTL01] There is no tricuspid stenosis.
☐ [TVSTL02] There is non significant tricuspid stenosis.
☐ [TVSTL03] There is significant tricuspid stenosis.

TV Vegetations & Abcess List

☐ [TVVA01] No vegetation is seen.
☐ [TVVA02] We cannot exclude vegetation at the tricuspid valve.
☐ [TVVA03] There is vegetation at
☐ [TVVA04] There is mobile vegetation at
☐ [TVVA05] There is abcess at

TV Repair / Prosthetic List

☐ [TVRPL01] Tricuspid valve post repair.
☐ [TVRPL02] Tricuspid valve post repair with ring annuloplasty.
☐ [TVRPL03] Tricuspid valve post replace with bioprosthetic.
☐ [TVRPL04] Tricuspid valve post replace with mechanical valve.
☐ [TVRPL05] There is no residual tricuspid regurgitation.
☐ [TVRPL06] There is …. residual tricuspid regurgitation.
☐ [TVRPL07] There is … residual tricuspid stenosis.
☐ [TVRPL08] Prosthetic valve is well-seated.
☐ [TVRPL09] Prosthetic valve opens well.
☐ [TVRPL10] Limited prosthetic valve is opening.
☐ [TVRPL11] Prosthetic valve is abnormal.
☐ [TVRPL12] The opening prosthetic valve is not well visualized.
☐ [TVRPL13] There is no thrombus see on prosthetic valve.
☐ [TVRPL14] There is possible thrombus on prosthetic valve.
☐ [TVRPL15] No leakage is seen.
☐ [TVRPL16] There is …… central leakage.
☐ [TVRPL17] There is …… paravalvar leakage.
☐ [TVRPL18] Doppler evaluation showed normal prosthetic tricuspid valve gradient.
☐ [TVRPL19] Doppler evaluation showed increased prosthetic tricuspid valve gradient.
☐ [TVRPL20] There is abnormal increase of prosthetic tricuspid gradient.
☐ [TVRPL21] The prosthetic vegetations are suspected.
☐ [TVRPL22] The prosthetic vegetations are present.
☐ [TVRPL23] The rocking suggests dehiscence in the tricuspid valve.

[TVAS01] Add Statement

☐

Result

Tricuspid valve is normal in structure and function. There is no tricuspid regurgitation.

Pulmonic Valve

☑ [PV01] Normal

Pulmonic valve appears normal. There is no pulmonary regurgitation.

PV Structure List

☐ [PVSL01] The pulmonic valve is not well visualized.
☐ [PVSL02] The pulmonic valve is grossly normal.
☐ [PVSL03] The pulmonic valve is thickened.
☐ [PVSL04] There is notching of PV Doppler flow consistent with increased pulmonary vascular resistance.
☐ [PVSL05] Absent pulmonary valve.

PV Function list

☐ [PVFL01] Pulmonic valve opens well.
☐ [PVFL02] There is doming of pulmonic valve with minimal stenosis.
☐ [PVFL03] There is doming of the pulmonic valve with restricted opening.
☐ [PVFL04] There is prolapse of
☐ [PVFL05] There is flail of

PV Regurgitation list

☐ [PVRL01] There is no pulmonary regurgitation.
☐ [PVRL02] There is trace pulmonary regurgitation.
☐ [PVRL03] There is mild pulmonary regurgitation.
☐ [PVRL04] There is mild-moderate pulmonary regurgitation.
☐ [PVRL05] There is moderate pulmonary regurgitation.
☐ [PVRL06] There is moderate-severe pulmonary regurgitation.
☐ [PVRL07] There is severe pulmonary regurgitation.

PV Stenosis list

☐ [PVSTL01] There is no pulmonary stenosis.
☐ [PVSTL02] There is non significant pulmonary stenosis.
☐ [PVSTL03] There is significant pulmonary stenosis with gradient of
☐ [PVSTL04] There is pulmonary stenosis.
☐ [PVSTL05] There is supravalvar pulmonary stenosis.

PV Vegetations & Abcess List

☐ [PVVA01] No vegetation is seen.
☐ [PVVA02] We cannot exclude vegetation at the pulmonic valve.
☐ [PVVA03] There is vegetation at
☐ [PVVA04] There is mobile vegetation at

PV Repair / Prosthetic List

☐ [PVRPL01] Pulmonic valve post repair.
☐ [PVRPL02] Pulmonic valve post replace with bioprosthetic.
☐ [PVRPL03] There is no residual pulmonary regurgitation.
☐ [PVRPL04] There is …. residual pulmonary regurgitation.
☐ [PVRPL05] There is … residual pulmonary stenosis.
☐ [PVRPL06] The prosthetic PV opens well.
☐ [PVRPL07] The prosthetic PV is opening limited.
☐ [PVRPL08] There is normal prosthetic PV gradient.
☐ [PVRPL09] There is abnormal prosthetic PV gradient.

[PVAS01] Add Statement

☐

Result

Pulmonic valve appears normal. There is no pulmonary regurgitation.

Mass / Thrombus

- [x] [MT01] There is no abnormal mass or thrombus.
- [ ] [MT02] Mass or Thrombus is cannot clearly visualized.
- [ ] [MT03] Spontaneous echo contrast is seen at
- [ ] [MT04] There is thrombus at
- [ ] [MT05] There is mass at
- [ ] [MT06] Myxoma is present at

There is no abnormal mass or thrombus.

[MT07] Add Statement

- [ ]

Result

There is no abnormal mass or thrombus.

Pericardium

- [x] [PERI01] Pericardium is normal.
- [ ] [PERI02] There is no pericardial effusion.
- [ ] [PERI03] There is pericardial effusion at
- [ ] [PERI04] No sign of tamponade is seen.
- [ ] [PERI05] Findings consistent with sign of tamponade.
- [ ] [PERI06] Pericardium is thickened.
- [ ] [PERI07] Pericardium is calcified.
- [ ] [PERI08] Findings consistent with constrictive pericarditis.

Pericardium is normal.

[PERI09] Add statement

- [ ]

Result

Pericardium is normal.

Aorta

- [ ] [AO01] There is no atherome plaque seen at decending aorta.
- [x] [AO02] There is atherome plaque at
- [ ] [AO03] There is aortic dissection at
- [ ] [AO04] Aortic diameter.

There is atherome plaque at descending aorta.

Add Statement

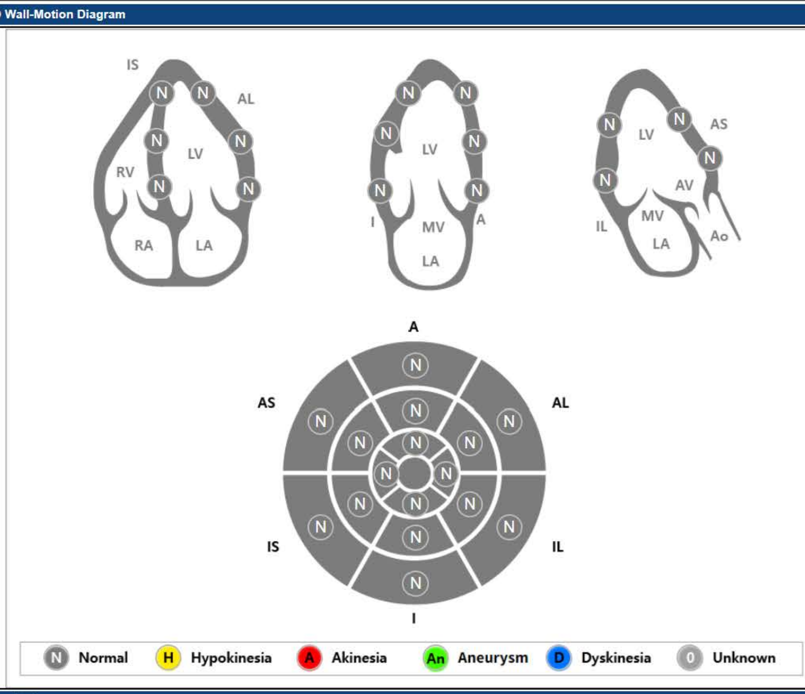
- [ ] [AO05]

Result

There is atherome plaque at descending aorta.

Others

2D Wall-Motion Diagram

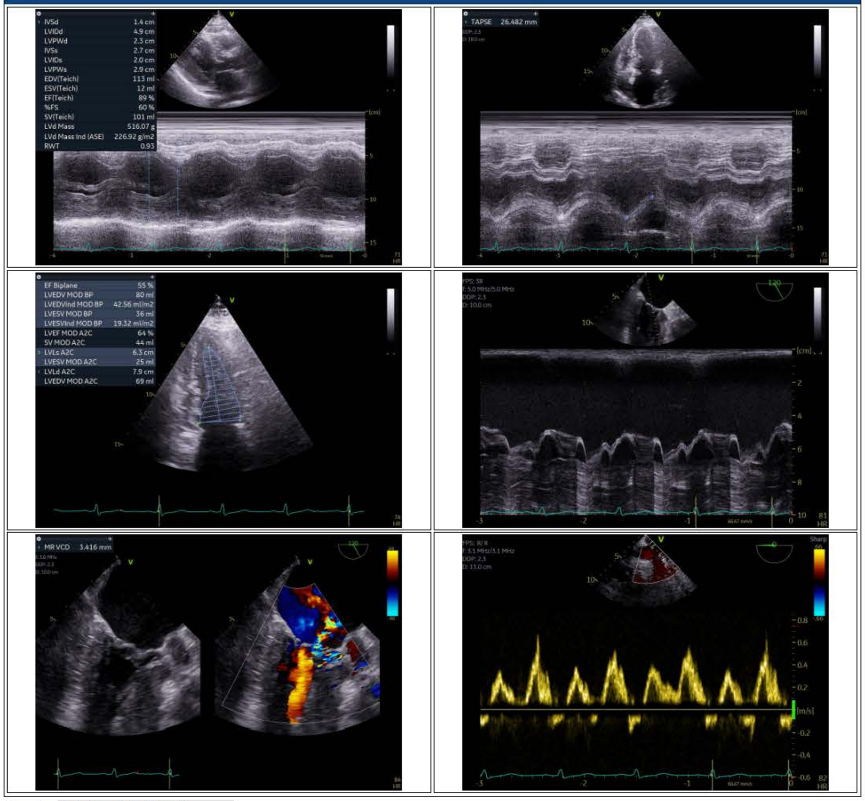


Conclusion

- Hyperthrophic Cardiomyopathy (HCM) with significant dynamic gradient at LVOT
- Moderate MR due to systolic arterior motion of AML
- Good LV systolic function, LVEF 55%
- Good RV contractility
- Diastolic function is indeterminate
- Global normokinetics

Comments

Picture



Operator : dr. Estu Rudiktyo, SpJP (K)

Fellow : GAT

Supervisor Confirm : dr. Estu Rudiktyo, Sp.JP

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